Department of Physics, FSK 116, Aegrotat Examination

PLEASE NEATLY WRITE / PRINT YOUR SURNAME AND NAMES AS REGISTERED IN GRADESCOPE!
(They will be read-in by machine!)

Surname and Names	Student Number

Examiners: Mr MJ Legodi
Moderator: Prof WE Meyer

Date: 8 July 2022
Time: 13:30 to 16:30

Total 91

FSK 116 OFFICIALS ONLY! Apply ✓ if Cheat-sheet is compliant, otherwise leave empty ☐

The following specific rules apply to this Examination

1. All test and examination regulations of the University of Pretoria are applicable.
2. Answer all questions and in the space provided on the question paper. If you need more space, use the "overflow pages" at the back of the question paper and clearly indicate where the question continues. Otherwise your work will not be marked!
3. Pocket calculators may be used, but not programmable calculators.
4. No cell phones or any other material are allowed in the test/examination venue.
5. This paper may not be taken out of the test/examination venue. It must be handed back to the chief invigilator, before you leave the venue.
6. Work written in pencil will not be marked. Write in black pen.
7. Provide labelled sketches, including co-ordinate systems, free-body diagrams, final answers with correct significant figures, etc. Failure to do so will result in marks deducted.
8. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. Only substitute numerical values in the final step of calculations! A correct answer does not mean that you will receive full marks.
9. Units must be used when numbers are substituted, else marks will be deducted.
10. Extra points will be awarded for high quality or elegant answers.

Constants, conversion factors and some formulae that may be useful

$g = 9.80\ \text{m/s}^2$ unless otherwise stated

Question 1 **[16]**

1.1 What mass of a material with density ρ is required to make a hollow spherical shell having inner radius r_1 and outer radius r_2? Suppose the difference in the radii cubes is given by: $(r_2^3 - r_1^3) = (1.5674 - 1.478)\ \text{m}^3$, calculate the mass to the correct number of significant figures. (3)

$\rho \equiv$ mass/volume

Mass $= \rho$ Volume

$= \frac{4}{3}\rho\pi(r_2^3 - r_1^3)$

$= \frac{4\rho}{3}\pi(1.5674 - 1.478)\ \text{m}^3$

3

$\approx \frac{4\rho\pi}{3}(0.0894)\ \text{m}^3$ 3 digits in bracketed value

OR

Only 2 significant!

$= 0.374477\rho\ \text{m}^3$ 4, π and 3 are known to an

$\approx 0.37\rho\ \text{kg}$ 2 sig fig infinite number of sig. Figures

where ρ has units kg/m^3 Explanation

1.2 A Warhol is a unit of fame or hype derived from Andy Warhol's dictum: "everyone will be world-famous for fifteen minutes". One Warhol represents, naturally, 15 minutes of fame. Some time ago, the average Tik Tok video was 15.6 seconds long before increasing to 60.0 seconds. How many Warhols do these average video lengths amount to? Give your answer using the correct prefix for your units. (3)

15 min = 1 Warhol

It is convenient to first develop a s $\leftrightarrow$ Warhol conversion.

$15\,\text{min}\left(\frac{60\,\text{s}}{1\,\text{min}}\right) = 1$ Warhol

$\Rightarrow 1\,\text{s} = \frac{1\ \text{Warhol}}{(15\times60)} = 1.1111\times10^{-3}$ Warhol

3

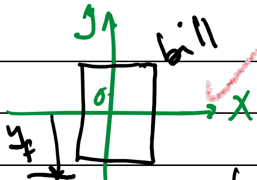
$\Rightarrow 15.6\,\text{s} = 15.6\,\text{s}\left(\frac{1.1111\times10^{-3}\ \text{Warhol}}{1\,\text{s}}\right) = 17.3$ mWarhol

And $60.0\,\text{s} = 60.0\,\text{s}\left(\frac{1.111\times10^{-3}\ \text{Warhol}}{1\,\text{s}}\right) = 66.7$ mWarhol

1.3 Floyd is playing a game with his friend, Julius. The aim of the game is for a player, Julius, to catch – and keep – a money bill as follows: Floyd is required to position the bill vertically, as shown in the figure, with the centre of the bill between but not touching Julius' index figure and thumb. Without warning, Floyd would release the bill. Julius has to catch the bill by pinching it between his fingers but without moving his hand downward. Assume that Julius, like other human beings, has an average reaction time of 200 ms. Will Julius catch the bill? Take the length of a R20 bill to be 136 mm. (5)

Floyd

Julius



Assume gravity acts on bill
In order for Julius to catch the bill, he h to snap on it before it has travelled a distance of half its length within the typical human reaction time.

How far will the bill travel in $t = 200$ ms?

5

$$y_f - y_i = v_{yi}t + \frac{1}{2}a_y t^2$$
$$y_f - 0 = 0 - \frac{1}{2}g t^2 = \frac{1}{2}(-9.80\,\text{m/s}^2)(200 \times 10^{-3}\,\text{s})^2$$
$$y_f = -196 \times 10^{-3}\,\text{m} \quad \text{(below } y=0 \text{ line)}$$

This length is $< \frac{163}{2}$ mm, ie half of the bill's length.

$\therefore$ Julius will not catch it.

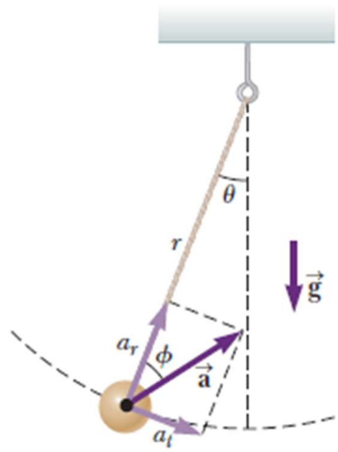
1.4 A particle on a frictionless surface has initial velocity (3.00 **î** + 6.00 **j**) m/s at $t = 0$ and an acceleration given by $\vec{a}(t) = (3.00\,t^2\,\hat{\mathbf{i}} + 2.00\,t\,\mathbf{j})\ \mathrm{m/s^2}$. Determine the magnitude of the particle's velocity at $t = 2.00$ s. (5)

$$\bar{v}(t=0) = 3.00\,\hat{i} + 6.00\,\hat{j}\ \text{m/s and } \bar{a}(t) = 3.00t^2\,\hat{i} + 2.00t\,\hat{j}\ \text{m/s}^2$$
$$\bar{v}(t) = \int \bar{a}(t)\,dt\ \text{m/s}$$
$$= \int(3.00t^2)\,dt\,\hat{i} + \int(2.00t)\,dt\,\hat{j}\ \text{m/s}$$
$$= [t^3\,\hat{i} + t^2\,\hat{j} + \bar{v}_{const}]\ \text{m/s where } \bar{v}_{const} = \bar{v}(t=0)$$
$$= [(2.00)^3\,\hat{i} + 3.00]\,\hat{i} + [(2.00)^2\,\hat{j} + 6.00\,\hat{j}]\ \text{m/s}$$
$$= 11.00\,\hat{i} + 10.00\,\hat{j}\ \text{m/s}$$

5

Question 2 **(25)**

2.1 A pendulum with a cord of length r = 1.00 m swings in a vertical plane (see figure). When the pendulum is in the two horizontal positions θ = 90° and θ = 270°, its speed is 5.15 m/s.



(a) Find the magnitude of the radial acceleration (in m/s^2) for these positions. (2)

(b) Find the magnitude of the tangential acceleration (in m/s^2) for these positions. (2)

(c) Draw a vector diagram to determine the direction of the total acceleration for these two positions. (2)

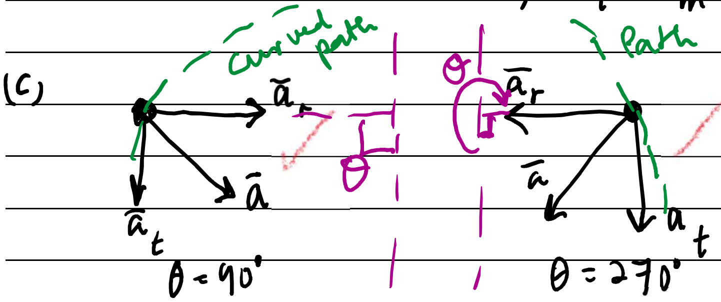
(d) Calculate the magnitude (in m/s^2) and direction (in degrees above or below the horizontal) of the total acceleration. (4)

Radial acceleration magnitude: $a_r = \frac{v^2}{r}$ (2)

(a) At both $\theta = 90°$ and $\theta = 270°$ positions $a_r = \frac{(5.15\ m/s)^2}{1.00\ m} = 26.5\ m/s^2$

(b) For both $\theta = 90°$ and $\theta = 270°$, $a_t = \frac{\Sigma F}{m} = \frac{mg\sin\theta}{m} = g\sin\theta = 9.80\ m/s^2$ (2)

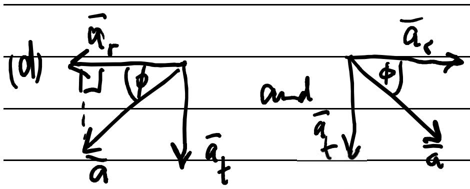
(c)



(2)

You may accept even if the angles are swapped.

(d)



$|\bar{a}| = \sqrt{a_r^2 + a_t^2} = \sqrt{[(5.15 m/s)^2]^2 + (9.80 m/s^2)^2} = 28.3\ m/s^2$

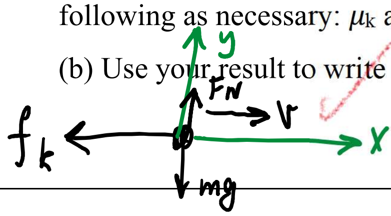
$= \sqrt{[(5.15 m/s)^2]^2 + (9.80 m/s^2)^2}$ (4)

$\phi = \tan^{-1}\left(\frac{a_t}{a_r}\right) = 20.3°$ below horizontal in both cases.

2.2 A puck resting on a sheet of ice is given an initial speed v_i in the $+x$-direction. The coefficient of kinetic friction between the ice and the puck is μ_k.

(a) Write an expression for the acceleration of the puck as it slides across the sheet of ice. (Use the following as necessary: μ_k and g. Indicate the direction with the sign of your answer.) (3)

(b) Use your result to write an expression for the distance d the puck slides, using v_i, μ_k, and g only. (2)



$$\bar{a} = \frac{\sum \bar{F}_x}{m} = -\frac{f_k}{m}\hat{i} = -\frac{\mu_k F_N}{m}\hat{i} = -\frac{\mu_k mg}{m}\hat{i} = -\mu_k g\,\hat{i}$$

(a) In y direction: $\sum F_y = 0 \Rightarrow F_N - mg = 0$
$\Rightarrow F_N = mg$

3

(b)

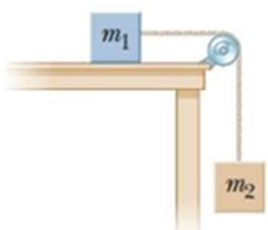
$$v_f^2 - v_i^2 = 2a_x(x_f - x_i) = 2a_x d$$

$$\therefore d = \frac{-v_i^2}{2(-\mu_k g)} = \frac{v_i^2}{2\mu_k g}$$

2

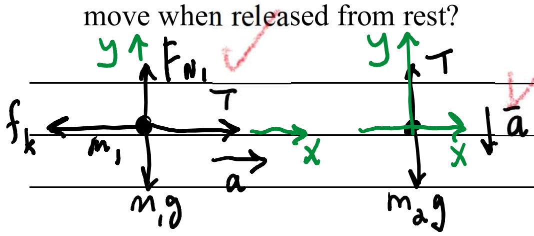
You may award 1 point if student used an expression containing t also.
eg. $(x_f - x_i) = d = v_i t + \frac{1}{2}a_x t^2$
with $a_x = \mu_k g$.

2.3 Consider the sketch below.



(a) Objects with masses m_1 = 12.0 kg and m_2 = 9.0 kg are connected by a light string that passes over a frictionless pulley as in the figure. If, when the system starts from rest, m_2 falls 1.00 m in 1.22 s, determine the coefficient of kinetic friction between m_1 and the table. (7)

(b) What would the minimum value of the coefficient of static friction need to be for the system not to move when released from rest? (3)



For m_2: $-T_2 + m_2 g = m_2 a_2 = m_2 a$
$-T + m_2 g = m_2 a$ ... ②

$\sum F_y = 0 \Rightarrow F_{N_1} = m_1 g$ for m_1
$\sum F_x = 0$ for m_2
Also, $\sum F_x = T_1 - f_k = m_1 a_1$ for m_1
$\Rightarrow T - \mu_k F_{N1} = m_1 a$ ... ①

We can write $a_1 = a_2 = a$ and $T_1 = T_2 = T$ because of the frictionless pulley and light ('massless') string.

Adding equations ① and ②:

$$m_2 g - \mu_k m_1 g = (m_1 + m_2) a$$

$$\therefore \mu_k = \frac{m_2 g - (m_1 + m_2) a}{m_1 g}$$

7

We can determine the acceleration from kinematics:

$$y_f - y_i = v_{yi} t + \tfrac{1}{2} a_y t^2 \Rightarrow a_y = \frac{2(y_f - y_i)}{t^2} = \frac{2(1.00\,m)}{(1.22\,s)^2} = 1.34372\,m/s^2$$

$$\Rightarrow \mu_k = \frac{m_2 g - (m_1 + m_2) a}{m_1 g}$$

$$= \frac{(12.0\,kg)(9.80\,m/s^2) - (9.0\,kg + 12.0\,kg)(1.34372\,m/s^2)}{(9.0\,kg)(9.80\,m/s^2)}$$

$$= 0.51$$

(b) $\Sigma F_x = m_1 a = 0$ and $\Sigma F_y = 0$

$\therefore \mu_s F_{N_1} = T$ $\qquad \therefore -T + m_2 g = 0$

$$\Rightarrow T = m_2 g$$

$$\Rightarrow \mu_s = \frac{T}{F_{N_1}} = \frac{m_2 g}{m_1 g} = \frac{9.0\,kg}{12.0\,kg} = 0.75$$

3

Question 3 **[15]**

3.1 Suppose that the co-ordinates of a particle undergoing motion under constant acceleration in the *xy*-plane are given by: $x_f = x_i + v_{xi}t + (1/2)a_x t^2$ and $y_f = y_i + v_{yi}t + (1/2)a_y t^2$. Show that the vector describing the position of the particle at time *t*, can be expressed as: $\vec{\mathbf{r}}_f = \vec{\mathbf{r}}_i + \vec{\mathbf{v}}_i t + (1/2)\vec{\mathbf{a}}t^2$, where the acceleration, $\vec{\mathbf{a}}$, is a constant. (5)

$\bar{r} = r_x \hat{i} + r_y \hat{j}$ where $r_x = x_f = x_i + v_{xi}t + \frac{1}{2}a_x t^2$

and $r_y = y_f = y_i + v_{yi}t + \frac{1}{2}a_y t^2$

$\Rightarrow \bar{r} = (x_i \hat{i} + v_{xi}t\hat{i} + \frac{1}{2}a_x t^2 \hat{i}) + (y_i\hat{j} + v_{yi}t\hat{j} + \frac{1}{2}a_y t^2\hat{j})$

5

Grouping vector components together:

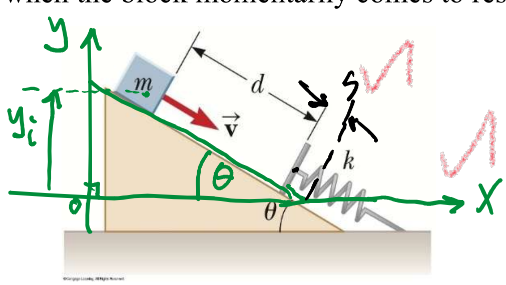
$\bar{r} = (x_i\hat{i} + y_i\hat{j}) + (v_{xi}t\hat{i} + v_{yi}t\hat{j}) + \frac{1}{2}(a_x\hat{i} + a_y\hat{j})t^2$

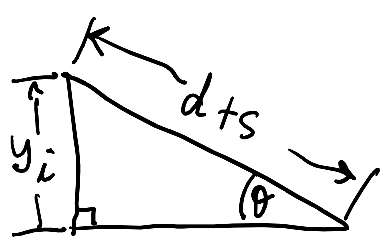
$= \bar{r}_i + \bar{v}_i t + \frac{1}{2}\bar{a}t^2$

3.2 Consider the figure below. An inclined plane set at angle θ has a spring of force constant k fastened securely at the bottom so that the spring is parallel to the inclined surface, as shown. A block of mass m is placed on the plane at a distance d. From this position, the block is projected/pushed downward toward the spring with initial speed v.

(a) From the above, develop an analytic expression for the distance, s, that the spring is compressed when the block momentarily comes to rest? Remember to draw a sketch and indicate the variables you will use and the assumptions you are making. (8)

(b) If, for the given figure, $\theta = 20.0°$, the spring constant $k = 500$ N/m, the block has mass $m = 2.50$ kg and it is initially located at a distance $d = 0.300$ m from the uncompressed spring and the block is projected downward toward the spring with speed $v = 0.750$ m/s. Determine the distance, s, that the spring is compressed when the block momentarily comes to rest? (2)





System: block, spring and Earth.

Set reference point for gravitational potential energy in the system at $y_f = y = 0$ as shown in sketch.

Mechanical energy is conserved in this isolated energy system

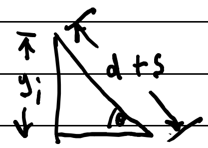
$$\Rightarrow \quad \Delta K + \Delta U = \Delta E_{mech} = 0$$

$$(K_f - K_i) + (U_{gf} - U_{gi}) + (U_{sf} - U_{si}) = 0$$

$$(0 - \tfrac{1}{2}mv_i^2) + (0 - mgy_i) + (\tfrac{1}{2}ks^2 - 0) = 0$$

$$\tfrac{1}{2}ks^2 = \tfrac{1}{2}mv_i^2 + mgy_i$$

$$\therefore \; s^2 = \frac{m}{k}v_i^2 + \frac{2m}{k}gy_i$$

from 

, we have $\sin\theta = y_i/(d+s)$

$$\Rightarrow \; y_i = (d+s)\sin\theta$$
$$= d\sin\theta + s\sin\theta$$

$$\Rightarrow \; s^2 = \frac{m}{k}v_i^2 + 2\frac{m}{k}gd\sin\theta + \left(\frac{2mg}{k}\sin\theta\right)s$$

8

$$0 = s^2 - \left(\frac{2mg}{k}\sin\theta\right)s - \left[\frac{2m}{k}gd\sin\theta + \frac{m}{k}v_i^2\right]$$

$$\therefore \; s = \frac{\left(\frac{2mg}{k}\sin\theta\right) \pm \sqrt{\left(\frac{2mg}{k}\sin\theta\right)^2 - 4\left[-\left(\frac{2mg}{k}d\sin\theta + \frac{m}{k}v_i^2\right)\right]}}{2}$$

OK

$$s = \frac{mg}{k}\sin\theta \pm \sqrt{\left(\frac{mg}{k}\sin\theta\right)^2 + \left(\frac{2mg}{k}d\sin\theta + \frac{m}{k}v_i^2\right)}$$

(b) $$s = \frac{(2.50\,kg)(9.80\,m/s^2)\sin 20.0^\circ}{500\,N/m} \pm \sqrt{\left(\frac{(2.50\,kg)(9.80\,m/s^2)\sin 20.0^\circ}{500\,N/m}\right)^2 + \cdots}$$

$$\cdots \; \frac{2(2.50\,kg)(9.80\,m/s^2)(0.300\,m)(\sin 20.0^\circ)}{500\,N/m} + \left(\frac{0.250\,kg}{500\,N/m}(0.750\,m/s)^2\right)$$

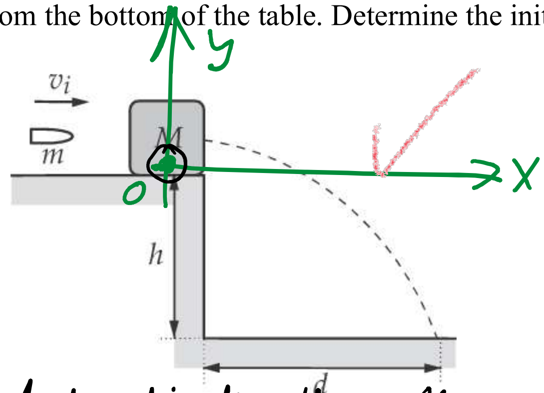
2

$$= 0.1198 \text{ or } -0.086$$

We reject the negative solution as unphysical

Question 4 **[9]**

A bullet of mass $m = 10.00$ g is fired into a block M of mass $M = 250$ g that is initially at rest at the edge of a table of height $h = 1.00$ m (see figure below). The bullet remains in the block, and after the impact the block lands $d = 2.00$ m from the bottom of the table. Determine the initial speed of the bullet. (9)



9

It is convenient to divide the problem into a linear momentum part and a kinematics part.

Momentum system: Bullet + block.

Perfect Inelastic collision since the bullet embeds into the blo

$\Delta p = 0 \Rightarrow p_{before} = p_{after}$ collision.

$m v_{1,i} + M v_{2,i} = (m+M) v_f$ where v_f is the velocity of $(m+M)$

$\therefore \quad v_{1,i} = \frac{(m+M)}{m} v_f$ since $v_{2,i} = 0$ for the block.

The bullet-block is a projectile with $a_x = 0$ and $a_y = -g = -9.80\,m/s^2$

$\therefore x_f - x_i = d = v_{xi} t = v_f t$

Also, since $v_{yi} = 0$, we can determine h from: $y_f - y_i = v_{yi} t + \frac{1}{2} a_y t^2$

ie. $y_f - y_i = -h - 0 = v_{yi} t + \frac{1}{2} a_y t^2 \Rightarrow t = \sqrt{\frac{2h}{g}}$

$$\Rightarrow v_f = \frac{d}{t} = \frac{d}{\sqrt{\frac{2h}{g}}} = \sqrt{\frac{gd^2}{2h}}$$

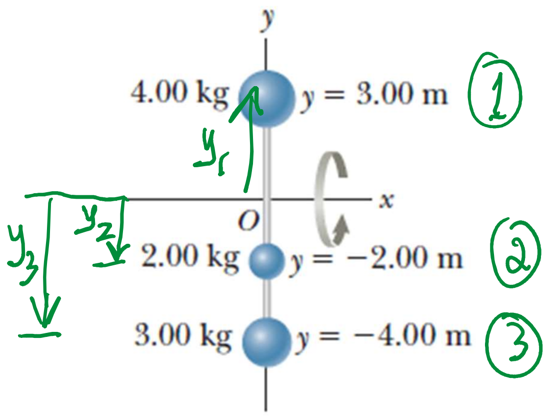
We are now in a position to calculate v_{1i}:

$$v_{1i} = \frac{(m+M)}{m} v_f = \underbrace{\left(\frac{m+M}{m}\right)\sqrt{\frac{gd^2}{2h}}}$$

$$= \left(\frac{10.0g + 250g}{10.0g}\right)\sqrt{\frac{(9.80\,m/s^2)(2.00\,m)^2}{2(1.00\,m)}} = 115\,m/s$$

Question 5 **[13]**

5.1 Consider the rigid rods of negligible mass lying along the y axis connecting three particles. The system rotates about the x axis with an angular speed of 1.50 rad/s.



(a) Find the moment of inertia about the x axis. (3)

(b) Find the total rotational kinetic energy of the system (2)

(c) Find the tangential speed of each particle. (2)

(d) Use your answer in (c) above to determine the total kinetic energy of the system. (2)

(a) $I_{Tot} = \sum I_i$ where I_i is the moment of inertia for the i-th mass. and is given by $I_i = m_i r_i^2$

$\therefore I_1 = m_1 y_1^2 = (4.00\,kg)(3.00\,m)^2 = 36.0\ kg\,m^2$
$I_2 = m_2 y_2^2 = (2.00\,kg)(-2.00\,m)^2 = 8.00\ kg\,m^2$
$I_3 = m_3 y_3^2 = (3.00\,kg)(-4.00\,m)^2 = 48.0\ kg\,m^2$
$\therefore I_{Total} = 92.0\ kg\,m^2$

(3)

(b) $K = \frac{1}{2} I \omega^2 = \frac{1}{2}(92.0\,kg\,m^2)(1.50\,rad/s)^2 = 104\ J$ (2)

(c) $V = \omega R$, where R is the radius of the particle's curved path.

$\therefore V_1 = (1.50\,rad/s)(3.00\,m) = 4.50\ m/s$
$V_2 = (1.50\,rad/s)(2.00\,m) = 3.00\ m/s$
$V_3 = (1.50\,rad/s)(4.00\,m) = 6.00\ m/s$

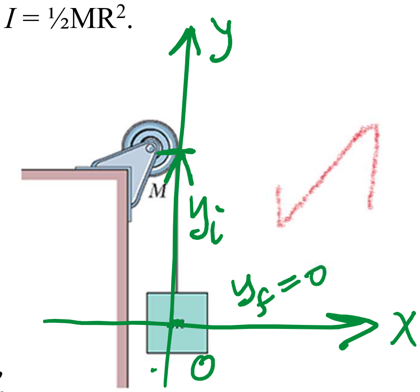
(2)

(d) $K_{Total} = \sum K_i = \frac{1}{2}(4.00\,kg)(4.50\,m/s)^2 + \frac{1}{2}(2.00\,kg)(3.00\,m/s^2) + \frac{1}{2}(3.00\,kg)(6.00\,m/s)^2$

$= 104\ J$

(2)

5.2 Consider the mass m = 1.250 kg suspended by a massless cord from a uniform disk with mass M = 2.500 kg and radius 0.300 m. The system has no friction and the system is released from rest. Use energy considerations to find the speed of the mass m after it dropped a distance of 1.000 m from the point of release. Take g = 9.800 m/s^2. ***HINT***: After dropping the 1.000 m, the block continues to fall and the disk is still turning! You need to evaluate the system constituents at two instances in time and consider their various types of energies when you apply the energy principles to the problem. Take the moment of inertia for the disk as $I = \frac{1}{2}MR^2$. (4)



System; Disk, block, Earth

Mechanical energy is conserved in the system, ie $\Delta E_{mech} = 0$

$$\Delta K + \Delta U_g = 0$$

$$(K_f - K_i) + (mgy_f - mgy_i) = 0$$

The Earth can be excluded in this equation since ΔK and ΔU, for Earth, are very small. Bonus point

Bonus point

$$\Rightarrow [(K_{f,disk} + K_{f,block}) - (K_{i,disk} + K_{i,block})] + (mgy_{f,block} - mgy_{i,block}) = 0$$

Since $\Delta U_{g,disk} = 0$ ie the disk does not translate, it only rotates.

We note that $\omega = v/R$

motion starts from rest!

$$\Rightarrow \left(\frac{1}{2} I_{disk}\,\omega^2_{f,disk} + \frac{1}{2} m v^2_{f,block}\right) - (0 + 0) + (0 - mgy_{i,block}) = 0$$

$$\frac{1}{2}\left[\frac{1}{2} MR^2 \left(\frac{v_{f,block}}{R}\right)^2\right] + \frac{1}{2} m v^2_{f,block} - mgy_{i,block} = 0$$

$$\frac{1}{4} MR^2 \cdot \frac{v^2_{f,block}}{R^2} + \frac{m}{2} v^2_{f,block} = mgy_{i,block}$$

4 Possible 6 points

$$v^2_{f,block}\left(\frac{1}{4}M + \frac{m}{2}\right) = mgy_{i,block}$$

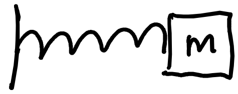
$$\Rightarrow v_{f,block} = \sqrt{\frac{mgy_{i,block}}{\left(\frac{M}{4} + \frac{m}{2}\right)}} = \sqrt{\frac{(1.250\text{ kg})(9.800\text{ m/s}^2)(+1.000\text{ m})}{\frac{2.500\text{ kg}}{4} + \frac{1.250\text{ kg}}{2}}} = 3.130\text{ m/s}$$

Question 6 **[13]**

6.1 A 300 g particle attached to a horizontal spring moves in simple harmonic motion with a period of 0.300 s. The total mechanical energy of the spring–particle system is 4.00 J.

(a) What is the spring constant (in N/m)? (2)

(b) What is the amplitude (in m) of the motion? (1)



$m = 300g$

$T = 0.300s$

(a) $\omega = \sqrt{\frac{k}{m}}$ and $T = \frac{2\pi}{\omega}$

$\Rightarrow T = 2\pi\sqrt{\frac{m}{k}}$ $\qquad E_{tot} = K + U = 4.00J$

$\therefore k = \frac{4\pi^2 m}{T^2} = \frac{4\pi^2(0.300kg)}{0.300s} = 132 \ N/m$ ②

(b) $E = \frac{1}{2}kA^2 \Rightarrow A = \sqrt{\frac{2E}{k}} = \sqrt{\frac{2(4.00J)}{132 N/m}} = 0.247 \ m$ ①

6.2 A transverse wave with an amplitude of 0.200 mm and a frequency of 430 Hz moves along a tightly stretched string with a speed of 1.96×10^4 cm/s.

(a) If the wave is modeled as $y = A\sin(kx - \omega t)$, what are A (in m), k (in rad/m), and ω (in rad/s)? (3)

(b) What is the tension in the string, if $\mu = 4.50$ g/m? (2)

(a) $A = 0.200mm = 0.200\times10^{-3}m = 2.00\times10^{-4}m$ (or $0.0002 m$)

$v = \lambda f \Rightarrow \lambda = \frac{v}{f} = \frac{1.96\times10^4 cm/s}{430 \ Hz} = 4.55\times10^{-3} m$ ③

But $k = 2\pi/\lambda = 2\pi/4.55\times10^{-3}m = 13.8 \ rad/m$

$\omega = 2\pi f = 2\pi(430 Hz) = 2.70177 rad/s = 2700 \ rad/s$

(b) $v = \sqrt{\frac{T}{\mu}} \Rightarrow T = \mu v^2 = (4.50\times10^{-3} kg/m)(1.96\times10^4\times10^{-2} m/s)^2$

$= 173 \ N$ ②

6.3 An aluminum calorimeter with a mass of 100 g contains 250 g of water. The calorimeter and water are in thermal equilibrium at 10.0 °C. Two metallic blocks are placed into the water. One is a 300.0 g piece of copper at 30.0 °C. The other has a mass of 70.0 g and is originally at a temperature of 100 °C. The entire system stabilizes at a final temperature of 20.0 °C.

(a) Determine the specific heat of the unknown sample. (4)

(b) Using the data in the table below, can you make a positive identification of the unknown material? (1)

The calorimeter, OR water, Cu and unknown metal constitute an isolated system.

(a) $\therefore \Delta Q = 0 \Rightarrow -Q_{hot} = Q_{cold}$; we use m to index unknown.

$$-(m_{Cu} c_{Cu} \Delta T_{Cu} + m_m c_m \Delta T_m) = (m_{Al} c_{Al} \Delta T_{Al}) + (m_w c_w \Delta T_w)$$

$$\therefore c_m = \frac{-(m_{Al} c_{Al} \Delta T_{Al} + m_w c_w \Delta T_w + m_{Cu} c_{Cu} \Delta T_{Cu})}{m_m \Delta T_m}$$

$$c_m = \frac{-[(0.100\,kg)(900\,J/kg\cdot°C)(20.0°C-10.0°C) + (0.250\,kg)(4186\,J/kg\cdot°C)(20.0°C-10.0°C)}{(0.070\,kg)(20.0°C-100°C)}$$

$$+ \frac{(0.300\,kg)(387\,J/kg\cdot°C)(20.0°C-30.0°C)]}{(0.070\,kg)(20.0°C-100°C)}$$

$$= 1.82214 \times 10^3\ J/kg\cdot°C \simeq 1822\ J/kg\cdot°C$$

Material	Mass (kg)	T_i (°C)	T_f (°C)
Al Cal	0.100	10	20.0°C
Water	0.250	10	20.0°C
Cu	0.300	30	20.0°C
Unknown	0.070	100	20.0°C

Bonus (Table)

(b) Material is likely Beryllium or an alloy of it (Not an exact identification, but very close!)

Specific Heats of Some Substances at 25°C and Atmospheric Pressure

Substance	Specific Heat (J/kg · °C)	Substance	Specific Heat (J/kg · °C)
Elemental solids		*Other solids*	
Aluminum	900	Brass	380
Beryllium	1 830	Glass	837
Cadmium	230	Ice (−5°C)	2 090
Copper	387	Marble	860
Germanium	322	Wood	1 700
Gold	129	*Liquids*	
Iron	448	Alcohol (ethyl)	2 400
Lead	128	Mercury	140
Silicon	703	Water (15°C)	4 186
Silver	234	*Gas*	
		Steam (100°C)	2 010

Note: To convert values to units of cal/g · °C, divide by 4 186.

END

OVERFLOW PAGES FOLLOW IN THE NEXT TWO PAGES!